

Prashant Choudhary

AI/ML Engineer | Python Backend | LLM & Agentic Systems

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LinkedIn | GitHub | Portfolio

Professional Summary

AI/ML Engineer and M.Sc. Natural Language Processing student at the University of Trier with professional experience building **Python/FastAPI backends** for public-facing AI products, **production RAG systems**, **agentic workflows**, and voice/IVR applications. Hands-on with PyTorch, Hugging Face, LangGraph, vector retrieval, model evaluation, and local-LLM infrastructure. Experienced across the full AI backend lifecycle, from data ingestion and retrieval to orchestration, inference, and production integration. Seeking Werkstudent roles in AI/ML, Python Backend, NLP, or Generative AI.

Education

University of Trier

M.Sc. Natural Language Processing

Trier, Germany

Apr 2026 – Expected Sep 2028

Uka Tarsadia University

B.Tech Artificial Intelligence & Data Science | CGPA: 7.26/10

Bardoli, India

2021 – 2025

Professional Experience

EnactOn Technologies / Proposal.biz

Junior AI/ML Backend Engineer

AI Prompt Engineer Intern

Surat, India

Jul 2025 – Nov 2025

Dec 2024 – Jul 2025

- Built **Python/FastAPI AI backends** for public-facing products, including **production RAG pipelines**, agentic workflows, and voice-based IVR systems for patient- and service-oriented applications.
- Contributed to the early AI backend of **Proposal.biz**, building a **website-to-vector knowledge pipeline** that crawled customer URLs, extracted and processed business content, and converted it into retrievable context for grounded proposal generation.
- Developed agentic proposal-generation workflows with **context/state management**, retrieval, web/search integration, and image-generation capabilities across multi-step document creation and editing.
- Integrated LLM providers, structured prompting, retrieval components, and guardrails into production services for reliable long-running AI workflows.

ALGOBRAIN AI

AI/ML Engineer Intern

Remote, India

Sep 2023 – Jan 2024

- Developed conversational-AI prototypes using **Python, TensorFlow, Rasa, and Dialogflow**, working on intent understanding, dialogue flows, and model/API integration.
- Built and evaluated ML-backed automation and chatbot workflows for applied customer-facing use cases.

Nothing

Community Product Tester

Remote

Feb 2023 – Present

- Tested pre-release software and hardware, documenting and reporting **150+ product issues and edge cases** to support release quality and stability.

Technical Skills

Programming & Backend: Python, FastAPI, REST APIs, SQL, PostgreSQL, SQLite, Supabase, Linux

Machine Learning & NLP: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Sentence Transformers, NLP, Deep Learning, Few-Shot Learning, Contrastive Learning, Model Evaluation

LLMs & Agentic AI: RAG, LangChain, LangGraph, CrewAI, Agno, Tool Calling, Prompt Engineering, Embeddings, Vector Search, Ollama, OpenAI/Gemini/Groq APIs

MLOps & Cloud: Docker, Git/GitHub, MLflow, AWS SageMaker, Google Vertex AI, Oracle OCI, Qdrant, ChromaDB

Selected AI/ML & Open-Source Projects

SetFit Reproduction & Paper-Code Audit Python, PyTorch, Hugging Face, Sentence Transformers *2026*

- Independently reimplemented SetFit few-shot text classification without the official SetFit package, including deterministic sampling, contrastive sentence-encoder fine-tuning, and downstream classification.
- Reproduced the paper's CR MPNet result across **10 independent runs at $88.61 \pm 1.61\%$ accuracy**, closely matching the published $88.5 \pm 1.9\%$.
- Identified and experimentally validated a discrepancy between the paper's documented learning rate and the effective released implementation through source-level investigation and controlled experiments.

PocketLLM Ecosystem Python, FastAPI, RAG, Ollama, Local LLMs, Supabase *2025 – Present*

- Built and maintain an open-source LLM ecosystem spanning cloud-connected and local-first assistants, including a **FastAPI backend** with provider abstraction, streaming inference, model discovery, encrypted BYOK credentials, and persistent conversations.
- Extended the ecosystem with **PocketLLM Lite**: local/offline inference, Ollama and GGUF model management, document ingestion with source-aware retrieval, tool calling, memories, Whisper transcription, and an OpenAI-compatible local server.
- Grew the main open-source PocketLLM project to **66+ GitHub stars and 11 forks**.

AetherMind / Forge-1 Python, PyTorch, Transformers, CUDA *2026*

- Designed a compact decoder-only language-model stack with a **~125M-parameter variant targeting 4 GB VRAM** consumer GPUs.
- Implemented RMSNorm, Grouped Query Attention, RoPE, SwiGLU, and weight tying, together with custom training, inference, and evaluation pipelines.
- Added memory-aware training with FP16 mixed precision, gradient checkpointing, and gradient accumulation for constrained hardware.

Moltjobs AI Product, LLM-Assisted Job Workflow *2026 – Present*

- Built an AI-assisted job-application platform with resume creation/import, AI-assisted tailoring, reviewable diffs, application tracking, follow-ups, interview preparation, and analytics-oriented workflows.
- Grew the product to **2,000+ users** while iterating on the end-to-end job-search and application workflow.

Achievements

- **Top 7 of 50 teams**, Amalthea Hackathon 2024, IIT Gandhinagar; built a multilingual AI healthcare solution with voice/IVR workflows using Python, FastAPI, Dialogflow, Twilio, Gemini, and PyTorch.
- Built open-source AI systems with measurable adoption, including **PocketLLM (66+ stars, 11 forks)** and **Moltjobs (2,000+ users)**.

Certifications

Oracle: OCI 2025 Data Science Professional; AI Vector Search Professional; OCI 2025 AI Foundations Associate

Machine Learning: Supervised Machine Learning: Regression and Classification, Coursera; High-Dimensional Data Analysis, HarvardX

Languages & Availability

English: Fluent, IELTS Academic 7.5 **German:** B1, actively improving **Hindi:** Native

Status: M.Sc. student in Germany; available for student-compatible employment | **Ready to relocate**